

Quiz 9

Instructions: You must show all of your work to receive credits for correct answers.

Problem 1. (6 points) Find the radius of convergence and the interval of convergence of the

power series $\sum_{n=1}^{+\infty} \frac{x^n}{n \cdot 3^n}$.

$$\begin{aligned}
 a_n &= \frac{x^n}{n \cdot 3^n} \rightarrow \lim_{n \rightarrow +\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow +\infty} \left| \frac{\frac{x^{n+1}}{(n+1) \cdot 3^{n+1}}}{\frac{x^n}{n \cdot 3^n}} \right| \\
 &= \lim_{n \rightarrow +\infty} \left| \frac{x^{n+1}}{(n+1) \cdot 3^{n+1}} \cdot \frac{n \cdot 3^n}{x^n} \right| = \lim_{n \rightarrow +\infty} \left| \frac{x}{3} \cdot \frac{n}{n+1} \right| \\
 &= \lim_{n \rightarrow +\infty} \left| \frac{x}{3} \cdot \frac{1}{1 + \frac{1}{n}} \right| = \frac{|x|}{3} < 1 \Rightarrow |x| < 3 \Rightarrow \underline{R=3}
 \end{aligned}$$

$$|x| < 3 \Leftrightarrow -3 < x < 3$$

When $x = -3$, we have $a_n = \frac{(-3)^n}{n \cdot 3^n} = \frac{(-1)^n}{n}$ and $\sum_{n=1}^{+\infty} \frac{(-1)^n}{n}$ converges

$x = 3$, we have $a_n = \frac{3^n}{n \cdot 3^n} = \frac{1}{n}$ and $\sum_{n=1}^{+\infty} \frac{1}{n}$ diverges

Thus the interval of convergence is $[-3, 3)$

Problem 2. (4 points) Find the radius of convergence of the power series $\sum_{n=0}^{+\infty} (-2)^n x^n$ and correspondingly the sum of this series as a function of x .

$$\begin{aligned}
 a_n &= (-2)^n x^n \rightarrow \lim_{n \rightarrow +\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow +\infty} \left| \frac{(-2)^{n+1} x^{n+1}}{(-2)^n x^n} \right| \\
 &= \lim_{n \rightarrow +\infty} 2|x| = 2|x| < 1 \Rightarrow |x| < \frac{1}{2} \\
 &\Rightarrow R = \frac{1}{2}
 \end{aligned}$$

For all $|x| < \frac{1}{2}$, we have

$$\sum_{n=0}^{+\infty} (-2)^n x^n = \sum_{n=0}^{+\infty} (-2x)^n = \frac{1}{1 - (-2x)} = \frac{1}{1 + 2x}$$